Tourist Safety Monitoring Using Rule-Based Anomaly Detection, Geofencing, and Alert Prioritization

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***Abstract* —**Tourism is a rapidly growing economic sector across the world with most of it's growth contributing from the remote areas, including forest areas, islands and untouched natural regions. These areas attract tourists in large quantities however monitoring tourists in such areas is a major unsolved problem. Traditional system becomes unreliable as manually monitoring such areas become time consuming. The paper proposes a three-stage method to detect movement anolmaly, use geo fencing to detect movement in potentially risky areas and reduce false positive alerts, by using a device as simple as a smartphone. The solution includes a client mobile application built using Kotlin to trace movement, a centralized control dashboard built using HTML 5, CSS3 and Javascript backed by a high-throughput Python FastAPI backend. The android application ingests the location dynamically with a rate of 5 seconds to less than 3 seconds in case of emergencies. The live anomaly detector identifies abnormal moving speed over various pre-defined factors for moving, prolonged immobility or continued signal/GPS loss.

Keywords — Smart Tourism, Tourist Safety, Anomaly Detection, Real-Time Monitoring, Geofencing, Machine Learning

I. INTRODUCTION

1.1 Background

Tourism is a sector that has shown significant improvements over the past few decades thereby becoming a highly contributing sector to a country's economy. Recent patterns indicate that tourists prefer remote, untouched natural regions that are mostly unexplored over crowded and famous tourist spots. However, ensuring tourist safety in such areas becomes a comes with challenges such as unstable internet connectivity, difficult terrain, unexpected weather patterns and unforeseen situations. Hence, it becomes essential to mitigate these challenges to ensure tourist-safety.

Traditional systems mainly rely on manual surveillance which comes with multiple limitations due to vast area coverage. These methods prove to be reactive rather than proactive which increases the risk of delays in case of emergency/critical situations. Modern approaches to solve this problem include use of GPS based tracking systems along with a platform to monitor and detect suspicious movements in real-time.

1.2 Problem Statement

Tourist safety becomes an important concern especially for remote/unexplored areas. The availability of location tracking applications can help in detecting presence but fails to address concerns such as – entry into hazardous/restricted areas, prolonged inactivity, unexpected route deviations, etc. However, relying completely on location tracking methodology limits the efficiency as movement anomalies alone may not truly indicate an emergency situation. Variations in geographic context around different terrains can affect the interpretation of anomalies which can lead to irrelevant results. Moreover, GPS inaccuracies, limited network connectivity can hinder the identification of threats to tourist's safety. Therefore, there is a need for an intelligent and context aware safety monitoring system that combines precise movement detection, geographic safety conditions, and alert filtering into a single coherent framework.

1.3 Objectives

The primary objective of this research is to propose a tourist-safety monitoring application along with a framework that can monitor multiple tourists simultaneously and provide timely and reliable alerts. The proposed solution aims to identify potentially risky situations by analyzing – unusual tourists movement patterns, anomalies or prolonged inactivity to detect suspicious patterns across remote areas. It also aims to reduce the false-positive alerts by filtering mechanisms to ensure that genuine emergency situations receive utmost priority.

II. LITERATURE REVIEW

Trajectory anomaly detection is a topic widely studied that explicitly isolates anomalies instead of profiles normal points. Liu et al. introduced the Isolation Forest algorithm, that performs significantly better than a near-linear time complexity distance-based method, ORCA, LOF and RF in terms of AUC and execution time, especially in large data sets.[1]. Shao et al. introduced a fine-grained framework for trajectory anomaly detection that addresses challenges that are associated with noisy trajectory data [2]. Similarly, Hu et al. investigated the context-aware trajectory anomaly detection by incorporating contextual information, such as points of interest, into the analysis of moving trajectories [3]. Cao and Akoglu proposed a method combining complementary detectors, so that different kinds of trajectory anomalies are captured instead of relying on a single detection rule [7]. Lee et al. introduced a partition-and-detect framework which splits trajectories into multiple segments and flags outlying sub-trajectories [8].

D'Amicantonio et al. presented uTRAND, an unsupervised method for detecting anomalies in traffic trajectories using semantic and topological information. The paper presents a methodological framework designed to address the challenges in anomaly detection within traffic intersections called uTRAND (Unsupervised TRajectory ANomaly Detector). It even allows to formulate simple rules to classify anomalous trajectories in a way suited for human interpretation. [4]. Geospatial monitoring and location tracking accuracy also remain the important considerations to develop safety-first monitoring systems. Özdemir and Tuğrul developed a real-time tracking and geofencing system using an Arduino Uno and Sim908 hardware setup paired along with a NodeJS and Leaflet.js web application. They proposed three error reduction techniques – Kalman filter, logistic regression and Moving Average filters which were evaluated against precise baseline measurements [5]. Breunig et al. proposed the Local Outlier Factor, which scores each point by comparing its local density with its neighboring points [10]. Saligrama and Zhao studied the local anomaly detection, which stated the relevance of identifying various observations within localized contexts [6]. Zhang et al. applied an isolation-based approach to trace GPS locations which reported high detection rates [9].

Together, these papers provide a foundation to develop the modern safety-first architecture for monitoring tourist movement followed by specific geo-fencing rules, along with reducing false positive alerts.

III. SYSTEM ARCHITECTURE

The proposed tourist-safety monitoring system is built on a modular combining movement anomaly detection, geospatial safety assessment, and alert refinement into a unified incident-response pipeline. As illustrated in Fig. 1, the architecture consists of an input and data-validation layer, a movement-analysis module, a geospatial context engine, an alert-refinement module, and an incident-response layer. A separate evaluation branch incorporates Isolation Forest as a baseline for comparative analysis. The architecture is designed to distinguish unusual movement from potentially safety-relevant events while retaining interpretable evidence for each generated alert.

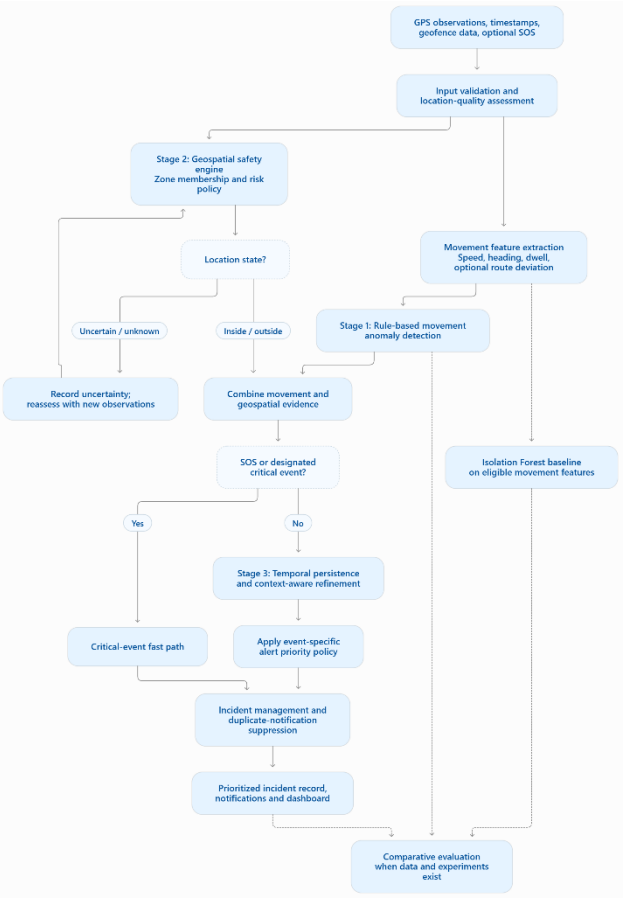


Fig.1. High-Level System Architecture Diagram

The various sections mentioned the above

system architecture diagram contribute to the pipeline as follows:

A. Input Layer: This layer works inside the android application built for the client. It works by collecting various parameters such as timestamped geographical observations, configured geofence definitions, and optional emergency signals such as SOS requests. It processes the tests and checks if any parameters violate the threshold definitions. Movement features are extracted from eligible observations, while unavailable information, such as a reference route, is handled explicitly

B. Movement Anomaly Detection: This module detects the patters that deviate from the normal observations by using a rule-based detector. It evaluates various parameters including speed, directional changes, dwell duration, and route deviation when a reference route is available. This modules provides anomaly indicators backed by supporting evidence.

C. Geospatial Safety Engine: This step evaluates the recorded dwelling observations against configured geographical areas ensuring they strictly adhere within the geo-fenced areas. It identifies potential risky conditions such as restricted-zone trespassing, unsual movement or uncertainty near geo fence boundaries. These conditions are evaluated independently.

D. Alert Refinement/Prioritization: This module evaluates each candidate events before sending notifications to the admin panel. It performs necessary checks for location quality handling, decision policies, etc.

E. Incident Response and Evaluation Layer: This layer works for the admin panel and dashboard. It converts the desired safety events into relevant type, assign priority, timestamp and design the outcome. This supports emergency escalation workflows ensuring reactiveness in real-time. An isolation forest model processes similar movement features and produces anomaly scores for side-by-side comparison with the rule based detection architecture.

The architecture builds a connected system that combines unsual movement, geo-fencing using rule-based risk evaluation mechanism and independent evaluation of various events.

IV. METHODOLOGY

The methodology proposed in this paper combines movement anomaly detection, geospatial assessment, and alert refinement to identify potentially risky tourist movements. The method evaluates movement observations independently of geographical zone membership before combining the resulting evidence. Isolation Forest model is evaluated separately as a baseline. The framework is assessed using labelled synthetic trajectories at both observation and event levels.

4.1 Problem Formulation and Dataset

Let the trajectory of tourist i be represented as an ordered sequence of GPS observations:

$$T_i = \{p_{i,1}, p_{i,2}, \dots, p_{i,n}\} \tag{1}$$

Each observation is defined as

$$p_{i,t} = (\phi_{i,t}, \lambda_{i,t}, \tau_{i,t}) \tag{2}$$

where $\phi_{i,t}$, $\lambda_{i,t}$, and $\tau_{i,t}$ denote latitude, longitude, and timestamp, respectively. Consecutive observations are recorded to derive various movement features and identify potential anomalies.

The geographical distance between consecutive observations is calculated using the Haversine formula:

$$\begin{aligned} a_t &= \sin^2\left(\frac{\Delta\phi_t}{2}\right) \\ &\quad + \cos(\phi_{t-1})\cos(\phi_t)\sin^2\left(\frac{\Delta\lambda_t}{2}\right), \\ d_t &= 2R\arcsin\left(\sqrt{a_t}\right) \end{aligned} \tag{3}$$

where $\Delta\phi_t = \phi_t - \phi_{t-1}$, $\Delta\lambda_t = \lambda_t - \lambda_{t-1}$, and R is the Earth's mean radius. Angular coordinates are expressed in radians. Given the elapsed time $\Delta\tau_t$, movement speed is estimated as

$$v_t = \frac{d_t}{\Delta\tau_t} \tag{4}$$

where elapsed time is measured in seconds and speed is expressed in metres per second. Each observation contains the latitude, longitude, and timestamp of the tourist. The observations that are invalid are considered as unavailable to reduce false positives.

A synthetic dataset was generated using random seed 42. It contains 289 synthetic observations, with six labelled anomaly events. It covers normal tourist movement, prolonged inactivity, abnormal or irregular speed, entry into prohibited areas, GPS boundary noise, and SOS-related observations.

4.2 Movement Anomaly Detection

In-order to identify deviations in tourist movement, the rule based detection is used in the first stage. The implemented rules examine factors like abnormal speed and prolonged inactivity. It checks for changes in direction or route deviation when the necessary route options/ reference route is available.

A rule for feature x_t can be represented as

$$A_t^{(x)} = \begin{cases} 1, & \text{if the configured anomaly} \\ & \text{condition is met,} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Here, $A_t^{(x)}$ indicates whether the observation satisfies the relevant rule. The applicable condition depends on the feature and its configured threshold. For example, speed is checked against an upper limit, while inactivity is assessed based on a time-based condition.

Movement anomalies are treated as indicators of concern instead of directly considering it as a sign of a tourist being in danger.

4.3 Geospatial Safety Assessment

The second stage evaluates observations against configured geographic zones. These zones are configured with a boundary and safety classification. For a circular zone with centre (ϕ_z, λ_z) and radius r_z, membership is defined as

$$G_t^{(z)} = \begin{cases} 1, & d(p_t, z) \le r_z, \\ 0, & d(p_t, z) > r_z, \end{cases} \quad (6)$$

where $d(p_t, z)$ is the geographical distance between the observation and the zone centre. A value of one indicates that the observation lies within the zone boundary and states that the tourists remains in the configured zone and thereby considered as safe.

The interpretation depends on the zone classification. Entry into a restricted zone can trigger a violation, whereas presence inside an ordinary zone does not by itself indicate risk. The GPS boundary noise and missing coordinates are treated as uncertain conditions rather than flagging them as critical emergency situations.

4.4 Alert Refinement and SOS Handling

The third applies checks to confirm if a recorded unsual activity continues long enough to be considered as suspicious. Repeated notifications associated with the same recorded event are prevented within the configured 5 minute cooldown period. Merging related alerts to cut false alarms has also been explored in intrusion detection systems [12], which motivates our refinement stage to reduce repeated alerts overload.

SOS events are configured to follow a priority path that can bypass the usual persistence checks. This ensures that genuine emergency situations are handled without any delay.

4.5 Evaluation Configurations and Metrics

The synthetic dataset compares the following four configurations:

1. **A — Movement rules only:** applies movement-based anomaly detection.
2. **B — Movement and geospatial assessment:** combines movement rules with geo fencing safety checks.
3. **C — Full refined pipeline:** combines movement detection, geospatial assessment, and temporal alert refinement.
4. **D — Isolation Forest:** provides a separate comparative baseline for anomaly detection.

At the observation level, the predicted labels are used to calculate precision, recall, F1-score, and false-positive rate. Precision measures the proportion of predicted anomalies that are labelled correctly. Recall measures the proportion of labelled anomalous observations that are detected. Their harmonic mean is

$$F_1 = 2\,\frac{P \cdot R}{P + R} \quad (7)$$

where P is precision and R is recall. The false-positive rate is calculated as

$$\mathrm{FPR} = \frac{FP}{FP + TN} \quad (8)$$

where FP and TN denote false-positive and true-negative observations.

Detection delay is measured from the beginning of a labelled event to its first compatible alert:

$$\Delta t_e = t_{\text{first alert}} - t_{\text{event start}} \quad (9)$$

The mean delay is calculated over detected events with valid timestamps. Point-level evaluation measures the individual detection accuracy, whereas event level evaluation measures whether the incident was detected and how early was the first alert was raised.

V. RESULTS AND DISCUSSION

5.1 Experimental Setup

The evaluation contains 289 synthetic observations accumulated over approximately 1.288 trajectory-hours, with six labelled anomaly events. Fig. 2 presents the distribution of the generated observation categories. Normal observations account for the highest - 207 points (71.6%). The remaining observations comprise prolonged inactivity (11), abnormal speed (2), restricted-zone entry (26), GPS boundary noise (30), an SOS-trigger point (1), and an SOS-active category (12).

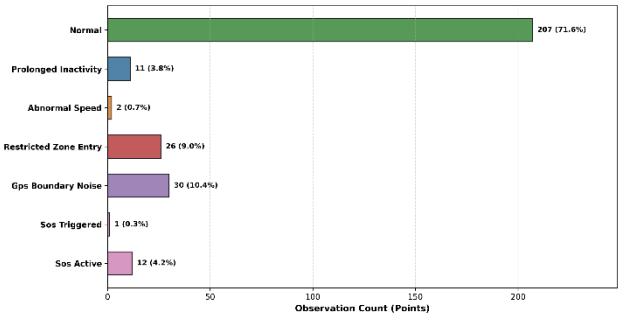


Fig. 2. Category Distribution (N = 289).

The four configurations were evaluated against the same generated data. Point-level metrics measure individual observations, while event-level metrics assess whether a relevant alert was generated for a labelled event. This distinction is important as the pipeline is designed to reduce the same alerts generated during an ongoing event.

5.2 Point-Level Detection Performance

Fig. 3 presents the comparison between the point-level metrics and the generated alert volume. Configuration B achieved the highest F1-score among the rule-based configurations, with precision of 0.8228, recall of 0.7927, and F1-score of 0.8075. Configuration A achieved higher precision (0.9286) but lower recall (0.3171), indicating that movement rules alone detected a smaller proportion of labelled anomalous observations. Isolation Forest achieved precision of 0.8448, recall of 0.5976, and F1-score of 0.7000.

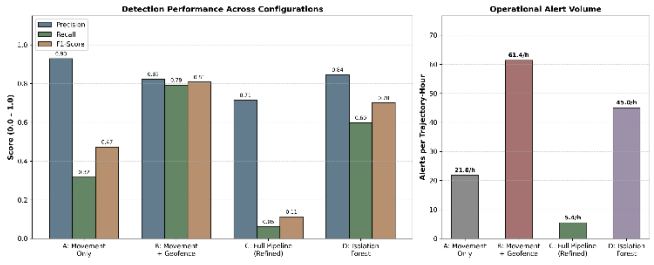


Fig. 3. Comparison of detection performance and alert volume.

Configuration C achieved precision of 0.7143, recall of 0.0610, and F1-score of 0.1124. Its low point-level recall reflects that it struggles to identify individual GPS observations as anomalous, even though it has the potential to detect an ongoing event.

Configuration C generates comparatively lesser alerts than Configuration B. However lesser alerts does not always ensure a better performance. It increases the rate of dropping true positives.

5.3 Event-Level Performance

At the event level, Configuration A detected two of the six anomalous incidents and thereby achieved a F1 Score of 0.4000. Configuration B detected three events and achieved an F1-score of 0.5000. Configuration C detected four events, with precision of 0.5714, recall of 0.6667, and F1-score of 0.6154. Isolation Forest detected 4 events and achieved F1-score of 0.6667, because it had higher precision.

Configuration C generated seven while Configuration B generated 79. Also, Configuration C failed to detect two incidents out of total 6 of them. In comparison with Isolation Forest, Configuration C took lesser time on average to generate an alert.

These results suggest the trade-off between reducing alerts and incident detection. Suppressing the repeated alerts can make the system less noisy, however, it increases the risk of genuine emergency incidents from being reported.

5.4 False Positives and Operational Trade-offs

Fig. 4 presents the point-level false-positive rate and the distribution of true-positive and false-positive observations. The first graph shows the false-positive rates, measuring how often the system incorrectly flags normal observations as anomalous. Configuration A and Configuration C both recorded a false-positive rate of 0.97%, compared with 6.76% for Configuration B and 4.35% for Configuration D. Configuration C produced fewer alerts than the other configurations, but its low point-level does not reflect it's reliability to detect the truly critical incidents.



Fig. 4. Point-level false-positive rate and distribution of true-positive and false-positive observations.

The second graph compares the number of true positives and false positives. It reflects that point-level and event-level findings are complementary rather than interchangeable. Both views are necessary to assess the balance between detecting incidents and avoiding redundant alerts.

5.5 Discussion and Future Work

The evaluated results highlight the importance of maintaining balance between detecting incidents and suppressing redundant alerts. The temporal alert refinement module reduces the number of similar alerts significantly, which suggests that the refinement mechanism can reduce notifications overload to the admin panel.

Future work aims to test the system with larger amount of tourist trajectories with wider variety of incidents to test its reliability on scale. It may also explore online learning-based detectors [11]. Such improvements can improve the understanding of the framework's effectiveness for different critical scenarios.

VI. CONCLUSION

This study presents a three-stage tourist-safety monitoring framework that combines rule based movement anomaly, geofencing assessment, and temporal alert refinement. The framework was evaluated against three alternative configurations with the help of 289 synthetic trajectories along with six labelled events.

The performance evaluation reflected that the configurations have different trade-offs between detection performance and alert volume. However the refined pipeline successfully detected four of the six labelled events and generated fewer alerts than the earlier configurations.

The future work focuses on testing the framework on larger and more diverse range of events with multiple parameters. These steps can help in determining whether the observed benefits extend beyond the simulated scenarios and support the framework's potential application in tourist-safety monitoring.

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